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Co-exposure to polystyrene microplastics and lead aggravated ovarian toxicity in female mice via the PERK/eIF2α signaling pathway

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A R T I C L E I N F O

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A B S T R A C T

Generally, individual microplastics (MPs) or lead (Pb) exposure could initiate ovarian toxicity. However, their combined effects on the ovary and its mechanism in mammals remained unclear. Female C57BL/6 mice were used in this study to investigate the combined ovarian toxicity of polystyrene MPs (PS-MPs, 0.1 mg/d/mouse) and Pb (1 g/L) for 28 days. Results showed that co-exposure to PS-MPs and Pb increased the accumulation of Pb in ovaries, the histopathological damage in ovaries and uterus, the serum malondialdehyde levels and decreased serum superoxide dismutase and sex hormone levels significantly when compared with single PS-MPs and Pb exposure. These observations indicated that co-exposure exerted more severe toxicity to mouse ovaries and uterus. Furthermore, co-exposure to PS-MPs and Pb caused endoplasmic reticulum (ER) stress by activating the PERK/eIF2α signaling pathway in the ovary, which resulted in apoptosis. However, the oxidative and ovarian damage were alleviated, and the mRNA levels of genes related to the PERK/eIF2α signaling pathway were down-regulated to levels of the control mice in the PS-MPs and Pb co-exposed mice administered with ER stress inhibitor (Salubrinal, Sal) or the antioxidant (N-acetyl-cysteine, NAC). In conclusion, our findings suggested that the combination of PS-MPs and Pb aggravated ovarian toxicity in mice by inducing oxidative stress and activating the PERK/eIF2α signaling pathway, thereby providing a basis for future studies into the combined toxic mechanism of PS-MPs and Pb in mammals.

1. Introduction

A sustained increase in plastic production has been observed over the past 70 years, and the annual plastic production is expected to continue to increase to approximately 590 million metric tons by 2050 (Statista, 2022). Due to poor recycling, million tons of plastics are accumulated annually in terrestrial and marine environments (Danso et al., 2019). Plastics in the environment can be decomposed by mechanical abrasion and ultraviolet radiation into microplastics (MPs), which are smaller particles with a 100 nm–5 mm diameter (Auta et al., 2017; Ng et al., 2018). MPs are also derived from intentional production for direct use or as precursors for other products, such as cosmetics, hand cleansers, and airblasting (Sharma and Chatterjee, 2017). MPs detected in the environment mainly include polystyrene, high/low-density polyethylene, polyethylene terephthalate, polypropylene, and polyvinyl chloride (Wagner et al., 2014). Polystyrene MPs (PS-MPs) are one of the most produced and typical MPs in the environment (de Sá et al., 2018). In 2015, the worldwide production of polystyrene represented 6.2% of all plastics demanded in Europe in 2018 (Hu et al., 2022). Xie et al. reported that PS-MPs abundance ranged from 328 to 82,276 particles m^{-2} were found in the Pearl River Estuary, China, and the PS-MPs in the size range of 1–2 mm were the most abundant, accounting for 38.0–57.4% of

Abbreviations: PS-MPs, Polystyrene microplastics; Pb, Lead; ER, Endoplasmic reticulum; Sal, Salubrinal; NAC, N-acetyl-cysteine; PERK, Protein kinase R-like ER kinase; eIF2α, Eukaryotic translation initiation factor 2α; ATF4, Activating transcription factor 4; CHOP, C/EBP homologous protein; HMs, Heavy metals; UPR, Unfolded protein response; FTIR, Fourier transform infrared spectroscopy; SEM, Scanning electron microscopy; EDS, Energy dispersive spectroscopy; H&E, Hematoxylin and eosin; E2, Estradiol; P, Progesterone; MDA, Malondialdehyde; SOD, Superoxide dismutase; RT-qPCR, Real-time polymerase chain reaction; IF, Immunofluorescence; IHC, Immunohistochemistry; Bip, Binding immunoglobulin protein; Bcl-2, B-cell lymphoma-2; Bax, Bcl-2 associated X protein.

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the entire PS-MPs (Xie et al., 2021).

Multitude studies have shown that MPs could accumulate in terrestrial animals, thereby inducing intestinal dysbacteriosis, inflammation, and metabolic disorder (Deng et al., 2017; Li et al., 2020). Previous studies have found that MPs could also harm the reproductive system. According to Hou et al. and Xie et al., exposure to MPs reduced the number of viable epididymal spermatozoa and increased sperm malformation in male mice (Hou et al., 2021a; Xie et al., 2020). Moreover, the ovary is a vital female reproductive organ. Research showed exposure to PS-MPs for 35 days could induce inflammation and reduce oocyte quality in the ovaries of mice (Liu et al., 2022).

MPs can adsorb other environmental pollutants, and the combined effect could potentially enhance their toxicity because of their large specific surface area and low surface polarity (Qiao et al., 2019; Wen et al., 2018). The existing study indicated that exposure to the MPs and heavy metals (HMs) might induce a higher pollution load of the marine medaka, resulting in empty follicles and follicular atresia (Yan et al., 2020). Lead (Pb) is one of the most frequently used HMs by humankind. Jiang et al. demonstrated that Pb impaired oocyte maturation and fertilization, which reduced the fertility of female mice (Jiang et al., 2021). Another study found that Pb exposure caused ovarian histopathological damages, as well as increased cell apoptosis (He et al., 2020). The affinity of Pb for plastic surfaces is relatively high, and MPs could be an effective vector of Pb (Davranche et al., 2019; Godoy et al., 2019; Yuan et al., 2020). Therefore, we hypothesized that the interaction between MPs and Pb might extremely affect the bioaccumulation of Pb and their combined toxicity. However, data about the combined ovarian toxicity of MPs and Pb to mammals are scarce, and further study is urgently needed.

Previous studies have revealed that oxidative stress and apoptosis are important molecular mechanisms that induce MPs and Pb toxicity (An et al., 2021; Ye et al., 2015). Interestingly, endoplasmic reticulum (ER) stress could be triggered by oxidative stress (Liu et al., 2021). The ER is an important organelle in mammalian cells and is the major location for protein synthesis, transport, and processing. ER stress occurs when certain factors cause dysregulation of calcium homeostasis and the buildup of misfolded and unfolded proteins in the ER lumen (Chen et al., 2014). At this time, cells initiate the unfolded protein response (UPR) to promote protein folding and degrade misfolded proteins, thereby reducing the burden and damage of the ER to some extent and restoring the protein homeostasis of the ER. However, prolonged ER stress could lead to reproductive system disorders through the malfunction of the UPR and increased cell apoptosis (Tang et al., 2019). One study reported that the reactive oxygen species (ROS)-regulated protein kinase R-like ER kinase (PERK)/eukaryotic translation initiation factor 2α (eIF2α)/C/EBP homologous protein (CHOP) signaling pathway was involved in the bisphenol A-induced reproductive toxicity (Yin et al., 2017). Another study showed that oxidative stress mediated the induction of ER stress and autophagy by microcystin-leucine arginine in KK-1 cells and mouse ovaries (Liu et al., 2018). Thus, ER stress mediated by ROS might be a crucial mechanism for the combined ovarian toxicity of MPs and Pb.

This study aimed to investigate whether PS-MPs affect the accumulation of Pb in female C57BL/6 mice. Meanwhile, histological analysis, biochemical analysis, immunoassay, and quantitative real-time polymerase chain reaction (RT-qPCR) analysis were conducted to investigate the effects and potential mechanisms of combined PS-MPs and Pb exposure on mice ovaries. This research will provide a better understanding of mammalian health risks associated with combined exposures.

2. Materials and methods

2.1. Characterization of PS-MPs and Pb

The PS-MPs dispersions (2.5% w/v, 10 mL) with diameters of 100 nm were obtained from Baseline ChromTech Research Center (Tianjin, China). These sizes of PS-MPs were chosen because small MPs are easier to transfer in the body and may cause more severe toxicity (Browne et al., 2008; Wu et al., 2022). Its size, surface charge and structure were analyzed by Zetasizer (Nano ZS90; Malvern Instruments Ltd., UK) and Nicolet iS50 Fourier transform infrared spectroscopy (FTIR, Thermos Scientific Co., Ltd., USA). Pb was prepared by dissolving Pb $(CH_3COO)_2$ $3H_2O$ powder (Xilong Scientific Co., Ltd., Guangdong, China) in an appropriate amount of ultrapure water. Two samples were prepared by dropping the PS-MPs (1 g/L PS-MPs) and PS-MPs + Pb (1 g/L PS-MPs + 2 g/L Pb) solutions onto the aluminized flakes and waiting for the solvent to dry naturally. Then, field emission scanning electron microscopy (SEM) was used to determine the morphology. Simultaneously, the elemental contents were determined by energy dispersive spectroscopy (EDS, JSM 6701F, JEOL Ltd., Peabody, MA, USA).

2.2. Experimental animals and design

SJA Laboratory Animal Co., Ltd (Hunan, China) provided us with sexually mature female C57BL/6 mice that were approximately 6 weeks old. Our animal experiments were approved by the Animal Care Review Committee of Nanchang University (approval number: 0064257) and had followed the Chinese guidelines for ethical review of experimental animal welfare. Prior to the experiment, mice were adapted to the laboratory animal room for one week, during which they were provided with clean drinking water and adequate food, relatively constant temperature and humidity, as well as a 12 h light/12 h dark cycle.

The mice were randomly assigned to four groups (n = 8 each group): (1) the control group; (2) PS-MPs group (0.1 mg/d/mouse PS-MPs); (3) Pb group (1 g/L Pb); and (4) PM group (0.1 mg/d/mouse PS-MPs + 1 g/L Pb). During the 28-day exposure period, the mice in group (1) were treated with uncontaminated physiological saline only. In groups (2) and (4), mice were gavaged with 200 μL of PS-MPs (0.5 g/L) once a day. The mice in groups (3) and (4) were given water that contained 1 g/L Pb ad libitum. The PS-MPs and Pb exposure doses were determined on the basis of previous reports (Deng et al., 2017; Togao et al., 2020; Xie et al., 2020), and the dose of PS-MPs was similar to environmentally realistic concentration of MPs in rivers ($\sim 10^6$ items m^{-3}) (Eerkes-Medrano et al., 2015). The mice were weighed regularly every morning. All mice were euthanized 1 day after the last gavage, and their organs and whole blood were gathered for later analysis.

2.3. Salubrinal (Sal) and N-acetyl-cysteine (NAC) supplementation

A total of 28 female mice were randomly assigned to four groups (n = 7 each group): (1) the control group; (2) the PM group; (3) PM with Sal (1.5 mg/kg) treatment group; and (4) PM with NAC (100 mg/kg) treatment group. Mice in group (1) were given physiological saline, whereas the three other groups of mice were given PM (0.1 mg/d/mouse PS-MPs + 1 g/L Pb) by gavage and drinking water. The mice in the group (3) were given Sal 30 min after the end of the gavage, and those in group (4) were given NAC by intraperitoneal injection once every 3 days. NAC and Sal were obtained from Beyotime Biotechnology Co., Ltd and MedChemexpress Biotechnology Co., Ltd, respectively. The doses of Sal and NAC were based on previous reports (Chen et al., 2020; Xie et al., 2020). At last, all mice were euthanized, with their organs gathered and weighed. The blood and organ samples were preserved at − 80 °C for subsequent analysis.

2.4. Pb element analysis

Approximately 20–40 mg of ovarian and uterine tissues were weighed and placed separately in a 1.5 mL centrifuge tube. In turn, 300 μL nitric acid and 100 μL perchloric acid were added to each centrifuge tube and then heated to 95 °C in water bath until the liquid was clear and transparent. The liquid was carefully sucked out from the tube and

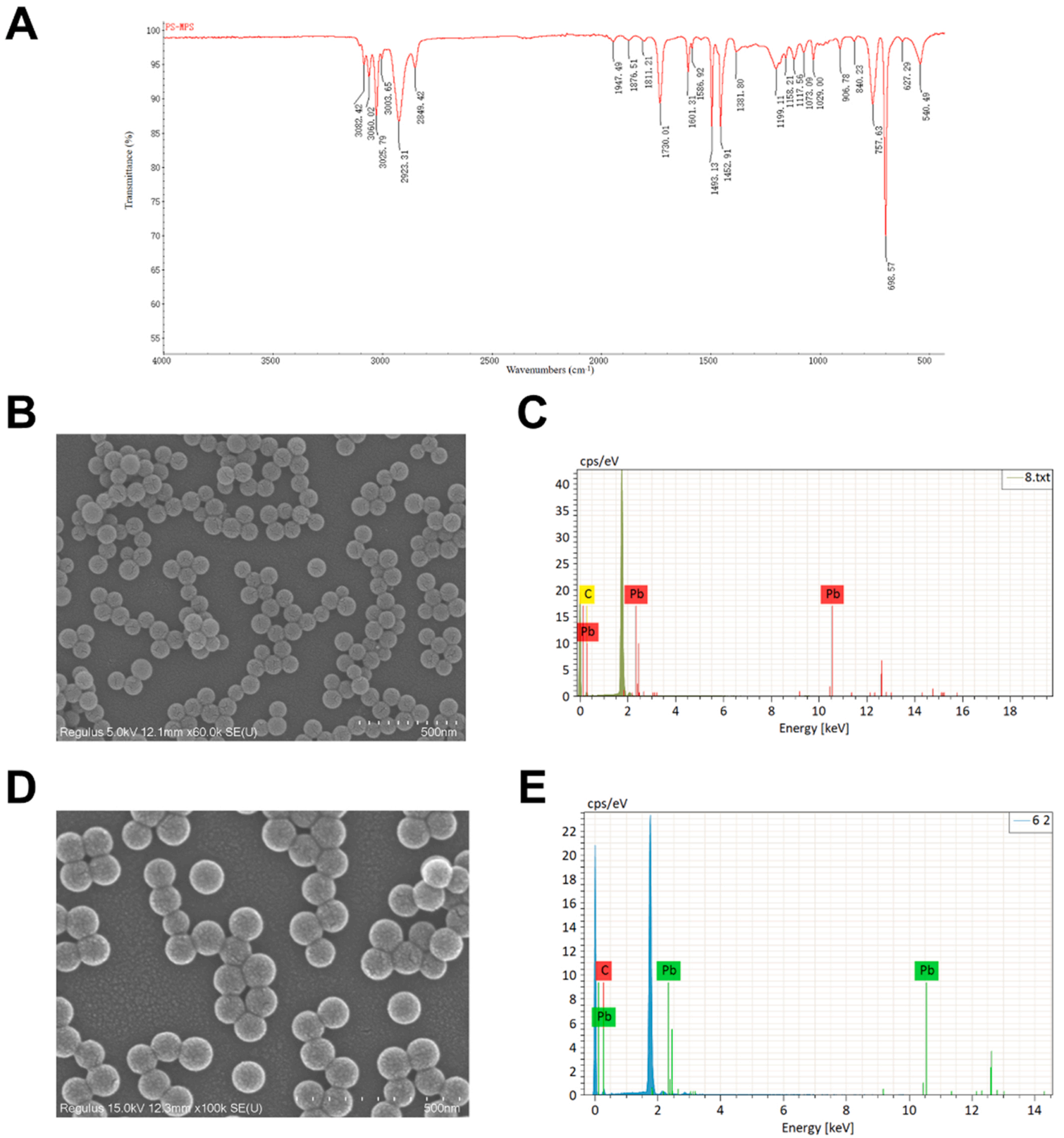

Fig. 1. Characterization of PS-MPs and Pb. (A) FTIR spectroscopy of PS-MPs. (B) SEM image of PS-MPs. (C) EDS analysis of PS-MPs. (D) SEM image of Pb-treated PS-MPs. (E) EDS analysis of Pb-treated PS-MPs.

diluted to 5 mL with ultrapure water. Pb concentration in the digestion solution was measured using inductively coupled plasma-mass spectrometry (ICP-MS, Varia 820-MS, US). Then, the contents of Pb in tissues were calculated according to the Pb concentration in the digestion solution and the weight of organs.

2.5. Histopathological

After separating the ovaries and uteruses from mice carefully, they were immersed in 4% paraformaldehyde solution immediately (n = 3 each group). Before being embedded in paraffin, the samples were dehydrated using different ethanol gradients. The sections of paraffin with a thickness of 5 mm were stained with hematoxylin and eosin (H&E). Subsequently, the stained sections were sealed using neutral gum. Finally, the Nikon T*i* optical microscope (Tokyo, Japan) was used to observe and photograph the sections. Histopathological changes in the ovary were evaluated based on the number of growing and atretic follicles, and histopathological changes in the uterus were evaluated based on endometrial thickness.

Endometrial thickness was quantified by measuring the diameter of

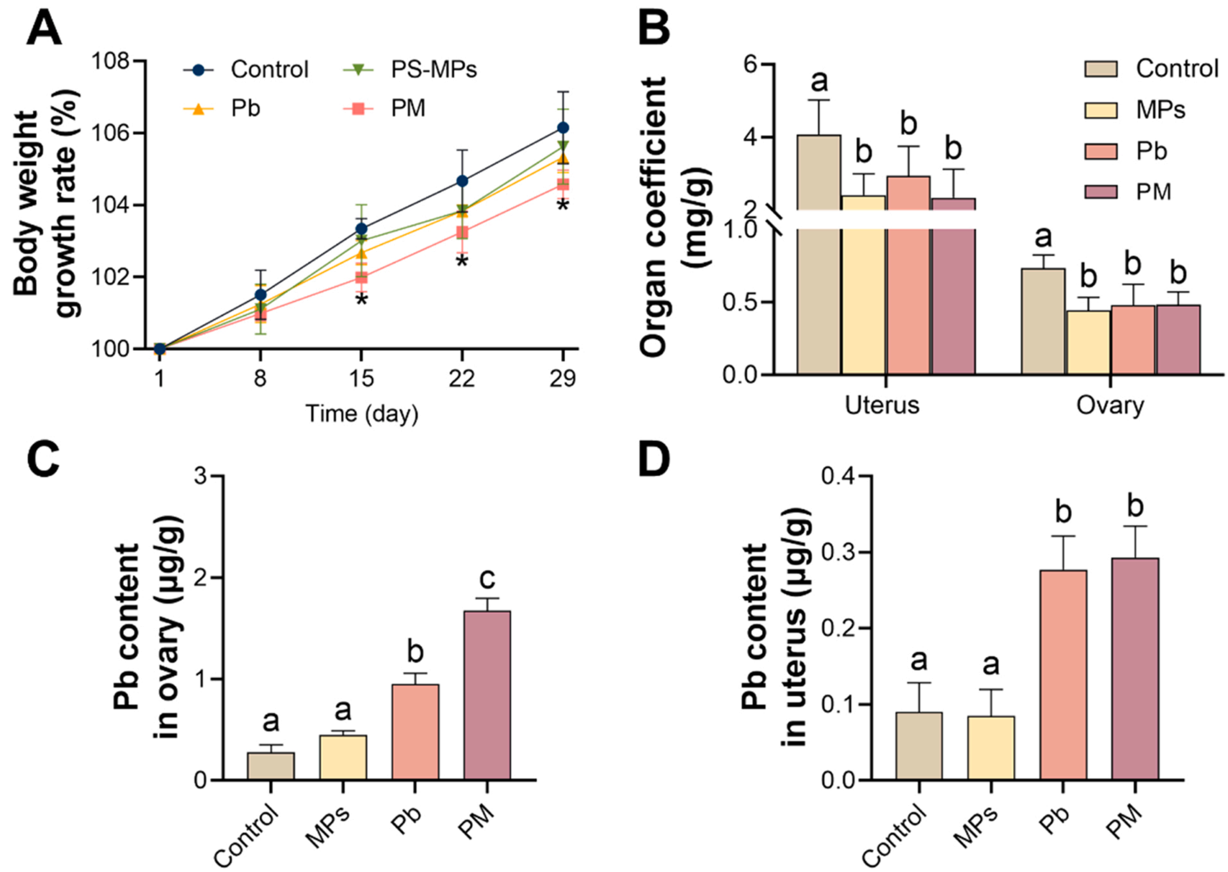

Fig. 2. Changes in body weight, organ coefficients and content of Pb element. (A) Body weight growth rate. (B) Organ coefficients of female mice. Content of Pb element in (C) ovary and (D) uterus. The presented values are mean ± SD. * indicates a significant difference compared with the control group ($P < 0.05$). Different letters (e.g., a, b and c) indicate a significant difference between the two groups ($P < 0.05$).

the endometrium using ImageJ 6.0 (National Institutes of Health, USA). Follicle stages were classified according to the previous definitions (Pascuali et al., 2018) with minor modifications and counted: (1) primary follicle: an oocyte surrounded by a single layer of cuboidal granulosa cells; (2) secondary follicle: 2 or more layers of cuboidal granulosa cells, without antrum; (3) antral follicle: follicle with an antral cavity, whatever the size. and (4) atretic follicle: with degenerating oocyte or pycnotic granulosa cells. Primary, secondary, and antral follicles were grouped as growing follicles.

2.6. Sex hormone analysis and oxidative stress analysis

After storing at 4 °C for 4–6 h, the collected blood samples were centrifuged at 4,000 for 10 min to separate the serum. Using ELISA kits (Shanghai YSRIBIO Industrial Co., Ltd., China) and following the instructions to evaluate the level of estradiol (E2) and progesterone (P). Appropriate amounts of serum were taken, and assay kits (Nanjing Jiancheng, China) were then used to measure the level of malondialdehyde (MDA) and the activity of superoxide dismutase (SOD) according to the laboratory manual.

2.7. RT-qPCR

Total RNA from ovaries was extracted according to the directions for users of the *Trelief* ™ RNAprep FastPure Tissue&Cell Kit (TSINGKE Biotechnology Co., Ltd., China), and the integrity and purity of the extracted RNA were verified. The extracted RNA with high purity and integrity was then reverse transcribed into cDNA using the Takara PrimeScript RT kit (Catalog#RR047A). The primer (Supplement Material Table S1, GENERAL Biosystems Co., Ltd., Anhui, China) and SYBR Green (TSINGKE, Catalog#TSE203) were mixed with the template DNA strand. Following the procedure used in a previous study (Tang et al., 2019), the Agilent AriaMx Real-Time PCR system (Agilent Technologies, Inc., Singapore) was used to perform RT-qPCR. Finally, *β-actin* was selected as an internal reference gene, and the $2^{-\Delta\Delta Ct}$ method was used to calculate the relative quantification of mRNA for each gene.

2.8. Immunofluorescence (IF) assay

The IF assay was performed to evaluate the expression levels of Binding immunoglobulin protein (Bip) and Caspase-12 in the ovaries. First, paraffin sections of ovarian tissue previously embedded in this study were deparaffinized using xylene and rehydrated with gradient ethanol. Next, the sections were subjected to antigen retrieval, serum blocking with 3% BSA, and incubation with Bip (1:500, Catalog#GB11098–1) and Caspase-12 (1:200, Catalog#GB111695). Then, the sections were subjected to incubation with enzyme-labeled secondary antibody, DAPI counterstain in nucleus, spontaneous fluorescence quenching, and cover slip with anti-fade mounting medium. Lastly, the sections were placed under a fluorescent microscope for observation and image acquisition. The mean pixel fluorescence intensity was measured using ImageJ 6.0.

2.9. Immunohistochemistry (IHC) assay

The IHC assay was performed to evaluate the expression levels of eIF2α and Cleaved-Caspase-3 in the ovaries. First, paraffin sections of ovarian tissue were deparaffinized and rehydrated. Then, the sections were subjected to antigen retrieval, endogenous peroxidase activity blocking, serum blocking, and incubation with eIF2α (1:500,

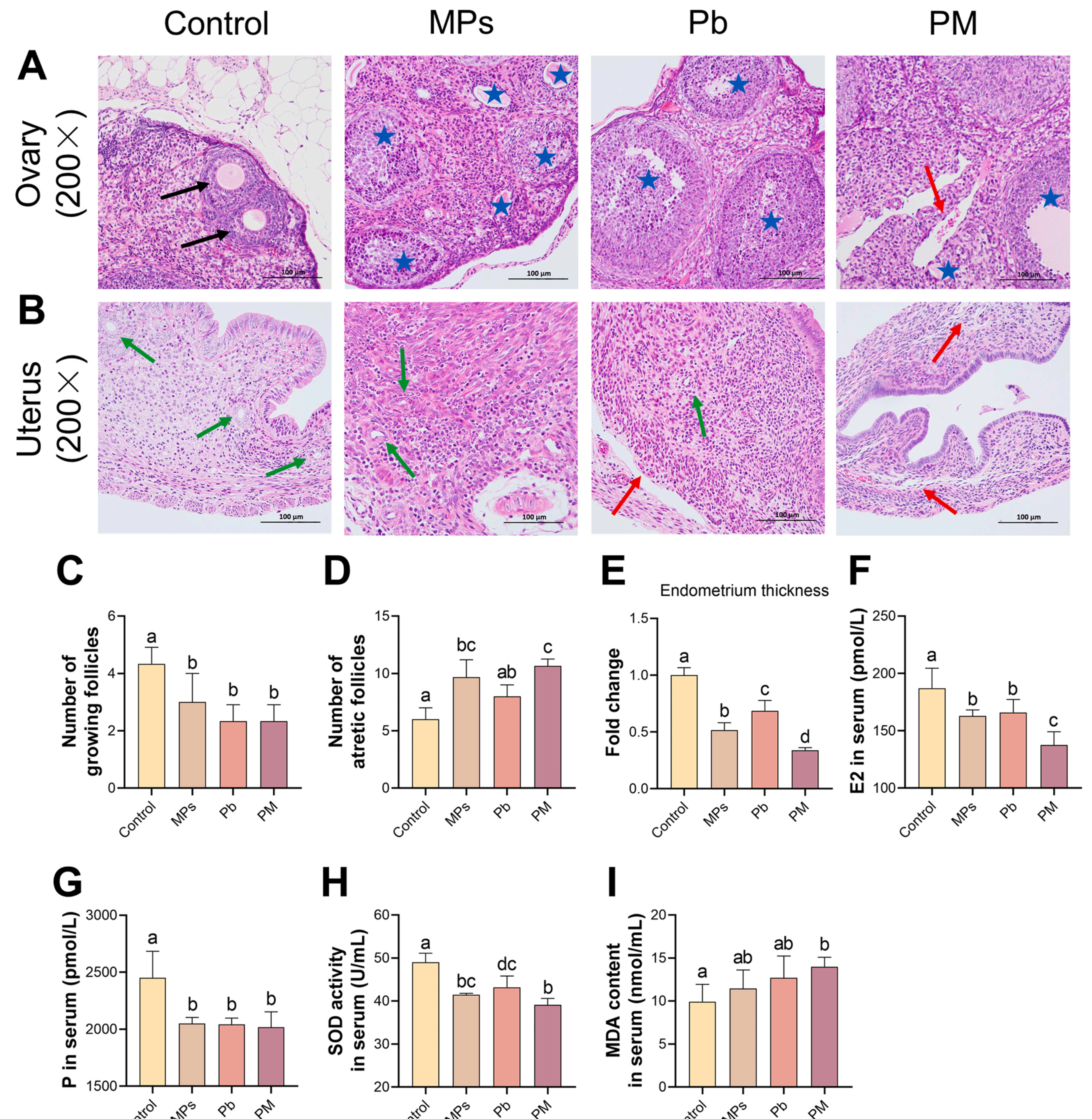

Fig. 3. Effects of PS-MPs and Pb on histomorphology, sex hormone and oxidative stress in serum. Sections of mice (A) ovary and (B) uterus with H&E staining (×200), black arrows indicate growing follicles; blue pentagrams indicate atretic follicles; red arrows indicate structure damage; and green arrows indicate uterine glands. (C) Number of growing follicles. (D) Number of atretic follicles. (E) Relative quantification of endometrium thickness. (F) Level of estradiol. (G) Level of progesterone. (H) Level of SOD. (I) Level of MDA. The presented values are mean ± SD. Different letters (e.g., a, b, c and d) indicate a significant difference between the two groups ($P < 0.05$).

Catalog#GB11544) and Cleaved-Caspase-3 (1:500, Catalog#GB11532). Next, the sections were subjected to incubation with secondary antibody, DAB chromogenic reaction, nucleus counterstaining and mounting medium. Lastly, all sections were imaged under an optical microscope. The immunohistochemical quantification was carried out by measuring the average optical density in 6 high-power fields of at least three mice ovaries by ImageJ 6.0. All reagents required for IF and IHC experiments were obtained from Wuhan Servicebio Technology Co., Ltd. (Hubei, China).

2.10. Statistical analysis

The data obtained from the experiments were processed to obtain the mean and standard deviation (expressed as mean ± SD). Data were compared among groups using the one-way analysis of variance (ANOVA) and the L.S.D. test in IBM SPSS Statistics 26 (SPSS, Inc.,

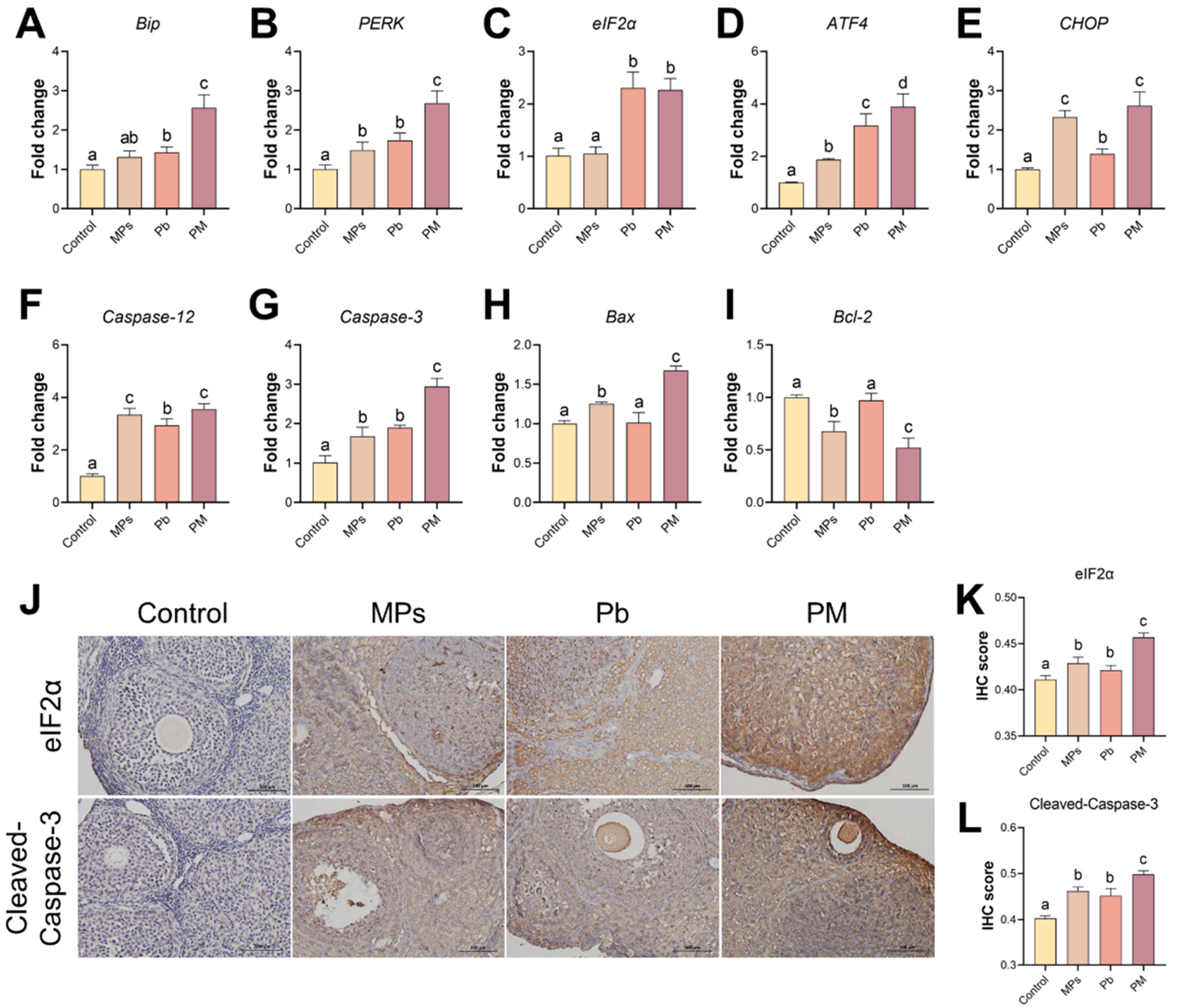

Fig. 4. Expression level of ER stress and apoptosis-related target factors. RT-qPCR results of (A) *Bip*, (B) *PERK*, (C) *eIF2α*, (D) *ATF4*, (E) *CHOP*, (F) *Caspase-12*, (G) *Caspase-3*, (H) *Bax* and (I) *Bcl-2*. (J) IHC images (×200) of eIF2α and Cleaved-Caspase-3. (K) IHC score of eIF2α. (E) IHC score of Cleaved-Caspase-3. The presented values are mean ± SD. Different letters (e.g., a, b, c and d) indicate a significant difference between the two groups ($P < 0.05$).

Chicago, Illinois). * indicates a significant difference compared with the control group ($P < 0.05$), and different letters (e.g., a, b, c and d) indicate a significant difference between the two groups ($P < 0.05$).

3. Results

3.1. Characterization of PS-MPs and Pb

The hydrodynamic size of PS-MPs in ultrapure water was approximately 100 nm and the zeta potential value of PS-MPs was around − 34 mV (Supplement Material Table S2). FITR analysis illustrated that the chemical composition of the plastic was polystyrene (Fig. 1A) according to previous studies (Jiang et al., 2020; Wu et al., 2019). The morphology of PS-MPs only and PS-MPs pre-mixed with Pb were observed by SEM. The SEM image showed that PS-MPs were uniform in size and had a regular spherical shape (Fig. 1B). According to the EDS result, the elemental content of Pb on the surface of PS-MPs was 0.71% (Fig. 1C, average of three replicates). SEM images of the PS-MPs compounded with Pb were similar to the PS-MPs only (Fig. 1D). Subsequently, they were scanned with EDS, and the results showed that the elemental content of Pb was 61.13% (Fig. 1F, average of three replicates). It was a remarkable increase in Pb content compared with PS-MPs without Pb treatment.

3.2. Effects of co-exposure to PS-MPs and Pb on body weights and organ coefficients

After 28 days of exposure, the body weight of mice in all groups increased to some extent. However, compared with the control group, the body weight growth rate of mice in the PM group decreased significantly from day 15 onwards, with no significant change in the PS-MPs and Pb groups (Fig. 2A). The outcomes of the organ coefficients of mice showed that the uterus and ovary coefficients were significantly lower in each group than those in the control group (Fig. 2B).

3.3. Accumulation of Pb in the ovary and uterus

According to the Pb content analysis, Pb concentrations in the mice

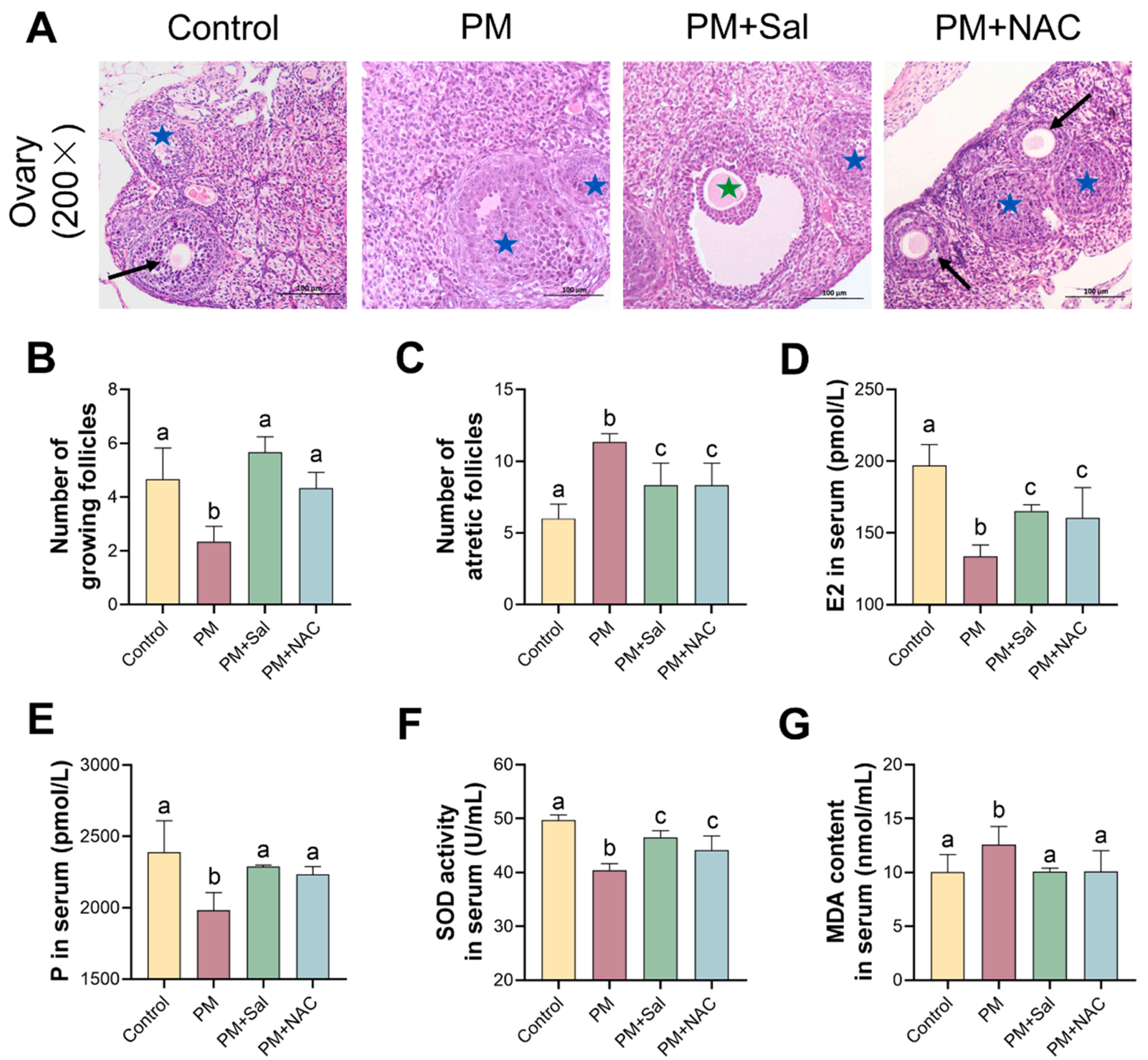

Fig. 5. Effects of Sal and NAC supplementation on the ovarian toxicity of mice in the PM group. (A) Sections of mice ovary with H&E staining (×200), black arrows indicate growing follicles; blue pentagrams indicate atretic follicles; and green pentagrams indicate mature follicles. (B) Number of growing follicles. (C) Number of atretic follicles. (D) Level of estradiol in serum. (E) Level of progesterone in serum. (F) Level of SOD in serum. (G) Level of MDA in serum. The presented values are mean ± SD. Different letters (e.g., a, b and c) indicate a significant difference between the two groups ($P < 0.05$).

ovary and uterus of Pb group and PM group have a significant increase. Meanwhile, no significant difference was observed in the PS-MPs and the control groups. Notably, the PM group had significantly higher Pb concentrations than the Pb group in the ovaries (Fig. 2C). In the uterus, the PM group had slightly higher Pb concentrations than the Pb group, although the distinction was not statistically significant (Fig. 2D).

3.4. Effects of co-exposure to PS-MPs and Pb on structures of the ovary and uterus, sex hormone, and oxidative damage

The results of the histopathological examination of ovaries showed a greater number of atretic follicles and fewer growing follicles in all exposure groups, and the ovarian tissue of mice in the PM group showed structural deficiency in comparison with the control group (Fig. 3A, C and D). In the uterus, mice in all exposed groups exhibited fewer uterine glands, a narrower glandular lumen, and a denser interstitium than the control group. Moreover, the PM mice had the thinnest endometrium, with the loss of glands and some structures of the lamina propria (Fig. 3B and E).

We evaluated the levels of estradiol, progesterone, SOD, and MDA in the serum to determine whether co-exposure to PS-MPs and Pb induces disturbances in sex hormone levels and oxidative damage (Fig. 3F–I). The results indicated that the levels of estradiol, progesterone, and SOD significantly decreased in all exposed groups in comparison with the control group, and the decrease degree of estradiol and SOD was more significant in the PM group of mice than those in the PS-MPs or Pb groups. A slight increase in serum MDA levels of PS-MPs and Pb groups were observed, whereas the PM group exhibited a significant increase in comparison with the control group.

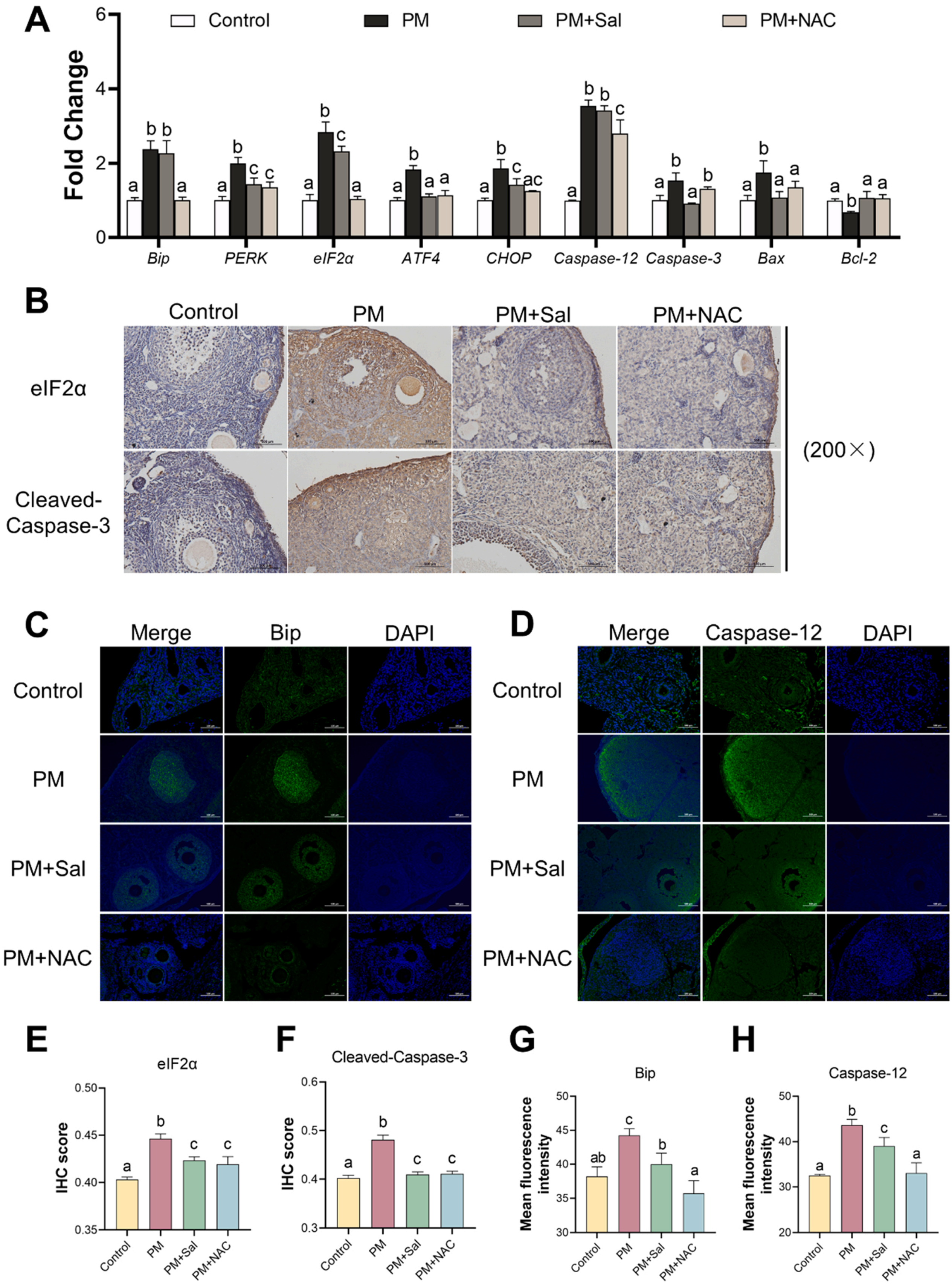

Fig. 6. Effects of Sal and NAC supplementation on the ovarian toxicity of mice in the PM group. (A) RT-qPCR results. (B) IHC images (×200) of eIF2α and Cleaved-Caspase-3. (C) IF images (×200) of Bip. (D) IF images (×200) of Caspase-12. (E) IHC score of eIF2α. (F) IHC score of Cleaved-Caspase-3. (G) IF score of Bip. (H) IF score of Caspase-12. The presented values are mean ± SD. Different letters (e.g., a, b and c) indicate a significant difference between the two groups ($P < 0.05$).

3.5. Co-exposure to PS-MPs and Pb activated PERK/eIF2α and apoptotic signaling pathways

To investigate the mechanisms of ovarian injury and dysfunction caused by co-exposure to PS-MPs and Pb in mice, the expression level of ER stress and apoptosis-related genes were measured using RT-qPCR (Fig. 4A–I). The results indicated that ER stress-related genes (e.g., *PERK*, *Bip*, *eIF2α*, *ATF4*, *CHOP*, and *Caspase-12*), as well as the pro-apoptotic genes (e.g., *Caspase-3* and *Bax*), were up-regulated significantly, whereas the anti-apoptotic gene (*Bcl-2*) was down-regulated significantly in the PM group in comparison with the control group. In addition, in the PM group, the changes in mRNA levels of the genes mentioned above were significantly different from those in the PS-MPs group or the Pb group. The quantification of IHC of eIF2α and Cleaved-Caspase-3 was significantly increased in the PM group in comparison with the MPs and Pb groups (Fig. 4J–L).

3.6. Sal and NAC prevented co-exposure to PS-MPs and Pb induced ovarian toxicity

The effects of the ER stress inhibitor Sal and the antioxidant NAC were used to further explore the mechanism of combined ovarian toxicity of PS-MPs and Pb in mice. Through the separate administration of Sal and NAC to mice co-exposed to PS-MPs and Pb, the ovarian histopathological damages were significantly alleviated in the PM+Sal and PM+NAC groups (Fig. 5A–C). When compared with the PM group, the levels of serum estradiol, progesterone, and SOD were significantly increased, whereas the level of MDA was significantly decreased in the Sal and NAC treated groups (Fig. 5D–G). Except for the *Bip* and *Caspase-12* genes, the mRNA levels of nearly all genes in the experimental group treated with Sal were restored to the control group. At the same time, the experimental group treated with NAC indicated that, except for the *Caspase-3* gene, the mRNA levels of nearly all genes were restored to the control group. The PM+NAC group had significantly lower levels of mRNA for the *Bip*, *eIF2α*, and *Caspase-12* genes than the PM+Sal group (Fig. 6A). Furthermore, the quantification of IHC of eIF2α and Cleaved-Caspase-3 was significantly decreased in the PM+Sal and PM+NAC groups compared with the PM group. The quantification of IF of Bip and Caspase-12 was also significantly decreased in the PM+Sal and PM+NAC groups in comparison with the PM group, with the decline being significantly greater in the NAC group than in the Sal group (Fig. 6B–H).

4. Discussion

MPs have been widely used in many applications, such as cosmetics, hand cleansers, and airblasting (Sharma and Chatterjee, 2017). Increasing studies have shown that MPs and Pb could cause ovarian toxicity (Liu et al., 2022; Massányi et al., 2020). However, studies on the combined toxicity of MPs and Pb and their internal mechanisms are still scarce. Therefore, in the present study, the female C57BL/6 mice were received to PS-MPs (0.1 mg/d/mouse) and Pb (1 g/L) separately and in combination for 28 days.

Analysis of body weight and organ coefficient in toxicology studies are important indicators in identifying potentially harmful effects of chemicals (Bailey et al., 2004). Our results indicated that the body weight growth rate and organ (ovary, uterus) coefficients decreased significantly in the PM group (Fig. 2A and B), suggesting that the ovary and uterus may be the target organs for PS-MPs and Pb-induced reproductive damage. In addition, this study showed that co-exposure to PS-MPs and Pb remarkably induced damage to ovarian and uterine structures, as demonstrated by an increase in the number of atretic follicles, a decrease in the number of growing follicles, and endometrial thinning (Fig. 3A–E). Follicles are the primary functional unit of the mammalian ovary, and ovarian granulosa cells generate sex hormones and different growth factors to sustain the development of the oocyte. Once the follicles are damaged, it can lead to sex hormone disorders (Hua et al., 2020). Early studies revealed that Pb could inhibit the synthesis of estradiol (He et al., 2020). PS-MPs could cause ovarian granulosa cells to apoptosis, thereby reducing ovarian reserve capacity (An et al., 2021; Hou et al., 2021b). Therefore, the most significant decrease in serum levels of estradiol and progesterone in the PM group (Fig. 3Fand G) might be explained by the fact that co-exposure to PS-MPs and Pb exacerbated their damage to the follicles.

To investigate the possible reasons for the greater ovarian and uterine toxicity of PS-MPs and Pb co-exposure in mice compared with their separate exposure, we measured the content of Pb by EDS in vitro (Fig. 1) and the Pb content in vivo (Fig. 2C and D). These results provided evidence that Pb was adsorbed on the surface of PS-MPs, and PS-MPs increased the accumulation of Pb in ovaries. Similarly, Lu et al. reported that MPs increased the accumulation of Cd in zebrafish tissues (Lu et al., 2018). Studies have shown that nanoparticles could penetrate the blood-testis and placental barriers, as well as the ovaries (Taylor et al., 2015; Wick et al., 2010). Therefore, we speculated that the sorption of HMs on MPs could transport metals into tissues (Brennecke et al., 2016), thereby increasing the accumulation and toxicity of Pb. However, our findings showed that PS-MPs did not increase the accumulation of Pb in the uterus, and more experiments are needed to explore the impact of PS-MPs and Pb co-exposure on reproduction and development in the future.

According to the literature, the main toxicity mechanism of PS-MPs and Pb is the induction of oxidative stress (Lee et al., 2019; Zhu et al., 2020). Oxidative stress results from an imbalance in the generation of ROS and the anti-oxidation system. The levels of SOD and MDA can reflect the body's ability to scavenge ROS and the extent of damage caused by ROS. According to our findings, in the PM group, SOD levels were significantly lower than those in the Pb group, whereas MDA levels were significantly higher than those in the control group (Fig. 3H and I), implying that co-exposure to PS-MPs and Pb caused more severe oxidative damage in mice. This result is similar to the finding of Qiao et al. (Qiao et al., 2019). However, the antioxidant NAC alleviated these changes (Fig. 5). These findings demonstrated that PS-MPs and Pb induced oxidative stress and impaired antioxidant capability. Except for oxidative stress, the ER stress signaling pathway is another key factor of concern. ROS was related to the induction of ER stress (Chen et al., 2014). Therefore, we hypothesized that ER stress might lead to ovarian toxicity. Bip is a major molecular chaperone protein of ER stress, and a considerable accumulation of misfolded proteins can induce a significant increase in Bip (Gardner et al., 2013). In our study, PS-MPs combined with Pb increased the transcription level of *Bip* in the ovary significantly (Fig. 4A), which suggested that the unfolded or misfolded proteins were increased after co-exposure to PS-MPs and Pb, resulting in ER stress. The three transmembrane proteins PERK, activating transcription factor 6 (ATF6), and inositol-requiring enzyme 1 (IRE1) on the ER membrane are separated from the chaperone protein Bip in response to ER stress, which activates the relevant signaling protein pathways downstream to enhance protein folding or remove misfolded proteins, thereby re-maintaining ER homeostasis (Senft and Ronai, 2015). However, sustained UPR activation induces apoptosis primarily through the activation of the PERK/CHOP pathway (Rutkowski et al., 2006). Activated PERK further activates activating transcription factor 4 (ATF4) and increases the expression of CHOP by phosphorylation of eIF2α (Walter and Ron, 2011). We observed that co-exposure to PS-MPs and Pb raised the transcription levels of *PERK*, *eIF2α*, *ATF4*, and *CHOP*, in addition to the translation level of eIF2α (Fig. 4B–E, J and K), which were similar to previous studies (Zhao et al., 2022; Zhu et al., 2021). This finding suggested that the combination of PS-MPs and Pb might activate ER stress through the PERK/eIF2α/CHOP signaling pathway.

CHOP is closely associated with the apoptosis induced by ER stress through down-regulating the expression of B-cell lymphoma-2 (*Bcl-2*), up-regulating the expression of Bcl-2 associated X protein (*Bax*) and activating Caspase-3 (Logue et al., 2013). Caspase-12 is another

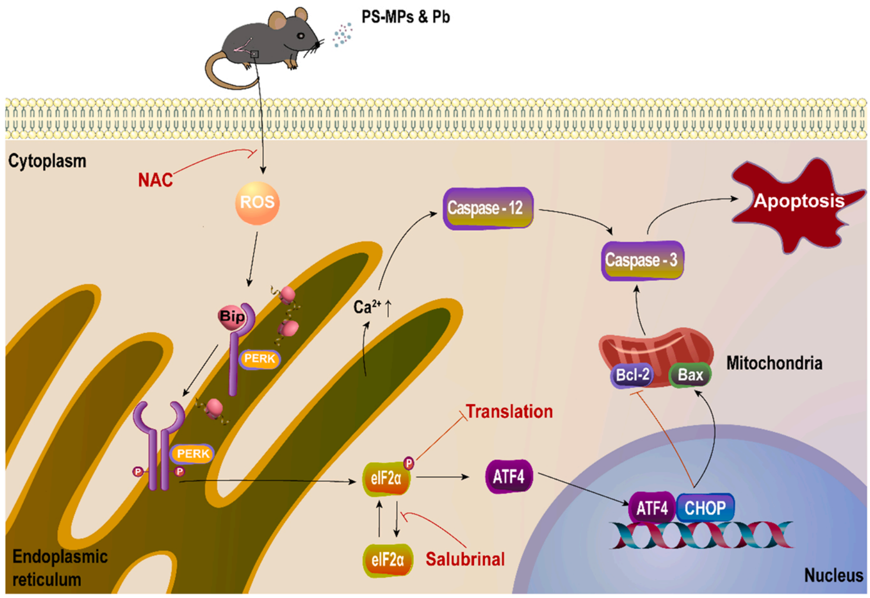

Fig. 7. Activation of the ER stress signaling pathway and the imbalance of Bax/Bcl-2 signaling induced by mitochondrial apoptosis in the ovaries of mice co-exposed to PS-MPs and Pb. "↑" indicates activation, "⊥" indicates inactivation, and "P" indicates phosphorylation.

apoptotic pathway induced by ER stress, which activates Caspase-9 and Caspase-3 (Szegezdi et al., 2003). In our study, co-exposure to PS-MPs and Pb increased mRNA levels of *Caspase-12*, *Bax* and *Caspase-3* genes, increased protein level of Cleaved-Caspase-3 and decreased mRNA level of *Bcl-2* gene (Fig. 4F–L). These findings suggested that the activation of ER stress might play a key role in ovarian toxicity induced by co-exposure to PS-MPs and Pb. In addition, the ER stress inhibitor Sal significantly attenuated the transcriptional activity of PERK/eIF2α and apoptosis signaling pathway-related genes, as well as the protein expression of Bip, eIF2α, Caspase-12 and Cleaved-Caspase-3 induced by the combination of PS-MPs and Pb (Fig. 6). These findings added to the evidence that the activation of the PERK/eIF2α signaling pathway in ER stress was a key factor in ovarian toxicity induced by co-exposure to PS-MPs and Pb. At last, we found that NAC treatment also significantly attenuated the activation of the PERK/eIF2α signaling pathway caused by the co-exposure to PS-MPs and Pb, which was similar to the findings of Liu et al., who confirmed that microcystin-leucine arginine induced oxidative stress in mouse ovaries and KK-1 cells occurring upstream of ER stress (Liu et al., 2018). Moreover, our results showed that NAC treatment inhibited the PERK/eIF2α signaling pathway more significantly than Sal treatment, suggesting that oxidative stress caused by PS-MPs and Pb co-exposure might occur upstream of ER stress, but more experiments are needed to explore it further. All results indicated that mice co-exposed to PS-MPs and Pb aggravated the apoptosis in the ovaries via oxidative stress and the activation of the PERK/eIF2α signaling pathway (Fig. 7).

Some limitations and issues should be further addressed in this study. The most urgent is to investigate whether MPs of different sizes induce discrepancies in ovarian toxicity. Then, more experiments are needed to explore the impact of PS-MPs combined with Pb on reproduction and development. Moreover, more protein expressions are needed to be evaluated to verify the toxicity mechanism in the future. The above considerations might contribute to a better understanding of the female reproductive toxicity induced by MPs combined with Pb and its mechanisms in future relevant studies.

5. Conclusion

In summary, our results indicated that the presence of PS-MPs increased the Pb bioaccumulation in mice, which potentially aggravated the combined toxicity of PS-MPs and Pb in female mice. Co-exposure to PS-MPs and Pb induced apoptosis through oxidative stress and the activation of the PERK/eIF2α signaling pathway, which ultimately led to ovarian damage. Moreover, the mechanism of ovarian toxicity caused by co-exposure of PS-MPs and Pb was confirmed by Sal and NAC supplement. Our findings contributed to further understand the combined toxicity of MPs and Pb, which would have important implications for mammalian reproductive health.

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CRediT authorship contribution statement

Yueying Feng: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Writing – original draft. **Hongbin Yua**: Conceptualization, Methodology, Investigation. **Wanzhen Wang**: Software, Validation, Data curation. **Yuanyuan Xu**: Formal analysis, Visualization, Data curation. **Jinfeng Zhang**: Investigation, Visualization. **Hengyi Xu**: Project administration, Funding acquisition, Writing – review & editing. **Fen Fu**: Resources, Project administration, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2022.113966.

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